

Social constraints (Discussion) ~ Avel Guénin-Carlut ~ ActInf Social Sciences

Courses/ActiveInferenceForTheSocialSciences | Avel Guénin-Carlut ~ ActInf Social Sciences 2023 | 52:40

Hello, welcome everyone. It is October 25th, 2023. We're in active inference for the social sciences. And today we're in a discussion section. It is Avil and I. And we'll start off by just saying hello and kind of reflecting on where we're at with the course. Then we have the questions form that is on the course website. It's in the video description. Feel free to submit questions via this form. And then today we're going to look at the questions that have been submitted and not answered yet. And we'll just give some thoughts on them. Also look for more questions to arise in the live chat. So with that, what do you want to begin with, Avil? Feel free to give any kind of overview or statement. And then we'll jump to the specific questions that have been submitted before today and during today.

The only statement here that I want to make is the last presentation I did. So prediction cause constraints to exist would be our wealth. Sorry of that, but I really, I don't have anything to say that is more specific right now.

All right. Well, this should be pretty fun. Let's jump into some questions. Okay.

What are the challenges and potential solutions in using active inference to model collective behaviors with various environmental constraints?

So the challenges first is that in any model, you have to pick what you include and what you did not include. So when you talk about boundary conditions, when you talk about what's including the model, you have to make hypothesis. Like if I tell you model, I don't know a country that is under sanctions like North Korea, Russia.

Okay. Okay. How do you model that? Do you make a set base for every possible products? Like every dimension accounts for a specific product or type of product?

And you have zero if you have access to it. And you have zero if you don't have access to it and one issue access to it. Okay. But what if we have new things that happen that, that, that, that is not very useful if you're interested in, is in innovation or something. So we'll have to make specific what is that, uh, we'll try to specific questions that you want to ask to the model and using active inference has.

And where it is relevant is that, uh, you can have, uh, uh, endogenous evolution of constraints using active inference. I think in a more meaningful sense than you'd have with a pure, let us say, heavy on or like non representational.

Uh, framework or framework that is presentational, but where the space presentation is like hard coded. So you, you, there is a space of, uh, models, let us say of evolution of certain norms, where I think active inference would have, um, uh, an edge because if inference is built to represent normativity and intentionality, and it is not the case to my knowledge of other, like either dynamicist or cognitivist, um, frameworks for, uh, or modeling.

Could you maybe restate and go into a little more detail? Was it non representational or indirect specification of what?

Of the, um, beliefs of the system.

So what's being directly specified, what's directly specified, and then what is being indirectly specified?

Inactive inference.

Inactive inference.

Yeah.

Inactive inference.

Inactive inference.

You have at least in a sense, uh, uh, correspondence between, uh, dynamical constraints, like information over where the system is and belief that the system holds that are akin to norms are by intentional states.

So beliefs or intentions are things that are going on inside.

And to my knowledge, there is no framework that makes, uh, an explicit.

For more connection between the two, you'd have dynamic framework, uh, where you have stuff that happens.

And there is no representation or intentional states.

And you have things like a utility theory where you have to specify, uh, preemptively what are the intentional states of the agents, which are, um, not very relevant to represent, um, like the analogous evolution of, uh, social constraints within a collective system.

So, um, um, situation where active inference would be useful is to model.

I am, uh, I will try to remember the term that Kaufmann used.

Kaufmann is filling and filling.

Um, in, um, in, um, so I will share the, can, can we both share the screen?

No, we can't.

You can add it directly in, in this, um, Coda page.

Oh, yeah.

Probably a little easier than just go to the backend course Coda page.

And then follow the token.

Yes, in there.

Yep.

And then go to the questions table and pop it out.

I don't have the questions that is on your screen.

Just click on this little expansion.

Um, uh, challenges and potential solutions and then pop it out.

Pop out what?

Um, just like next to the row C, this expand row.

And then you'll be in the same place.

Row C?

What is the row C?

Just there is no, no, no, no, no.

Just look visually at the expand row.

Look visually at the expand row.

Just click the expand row.

On the question that we're addressing and you'll pop into this campus.

The question that we are addressing is not on the page, at least on my terminal.

So I can't click.

Are you looking at the public front end or at the course backend?

I'm looking at the backend.

Okay.

And I do not have a question.

What are blah, blah, blah.

Oh, it is a sub question.

What are the challenges?

Okay.

Okay, cool.

All right.

Fun.

You were talking about Kaufman though.

What was it?

Yes.

It's a paper that he co-signed.

And who's my author is a typophylline, a critic of economic, like I say rationality, but no, they say it's a boring rationality.

So the dictionary of cognitive biases approach to modeling human behavior.

And basically the point of the paper, as I understand it, is that you have an evolution of the goals and norms that shape a social ecology and this kind of open-ended evolution of the intentional states of the agent, I think is, can be addressed in a more meaningful way with active inference than with any other formal framework kind of.

There are several other aspects, I think.

So first to kind of restate the challenges and then go to the potential solutions and the advantages, like the advantages in principle and in practice.

So you started with identifying like the major, any kind of modeling challenges.

So active inherits from all of these modeling challenges or areas that are difficult to step aside from.

And then depending on the resources of the group and whether it's 2023 or 2000, you know, future time when there's different tooling.

So that is just about the challenge of active inference sometimes compared today with more off the shelf approaches, which also may or may not work.

But in an act-inf you'll be spending your time making something that may or may not work, may or may not be orders of magnitude more just sophistication of a modeling setting than the team is ready to.

Again, that's a general modeling point.

Again, that's a general modeling point, but it's also very easy to get into with act-inf and to jump the gap between something that is pedagogical and helps show something like intuitive or cool.

And making something for example, that can be used in high reliability settings.

And that's a jump that hasn't been necessarily conveyed in the open source setting.

So it's difficult to see examples of, whereas there's kind of like production type machine learning algorithms that can at least be.

No, that's just a little general point.

So what are some advantages?

There's some really interesting concepts related to this, like indirect specification.

So like, let's just say that we were talking about the mountain car example or a social variable that seemed to be moving within a certain range.

Then the direct encoding of that constraints would be basically made outdated.

Okay, now the landscape changes.

And then all of a sudden the mountain car can go twice as far to a different direction.

So if that constraint had been explicitly encoded about cognitive variables or about the x position of the mountain car, then the model goes stale.

But if what's encoded is somehow a relation with the agent and the environment, so the mountain car has 10 horsepower and it can go left or right as its affordances and everything that builds from there.

And then here's the laws of physics or mechanics such that here's the weight of the car.

And then if the landscape changes or is modeled as changing, then you get, oh yeah, now it can go further or shorter in this direction.

But that's like an outcome of the relationship between the agent and the environment explicitly represented as such versus just looking at the outcome and then putting constraints on the outcome.

Yes.

To say we have historically the concept of explanation that we have inherited from the 19th, 20th century is that we explained by having a lawful relationship between variables.

And to do that, we have to pre-specify what are the relevant variables.

And this is not a thing that exists with open-ended evolutionary dynamics.

This is all in the Genna-Karlou, cognitive agency, something, something paper.

So, and so when we ask questions to a model, we don't want to have the answer, yeah, it will evolve and interestingly, we don't really know.

This is not what is expected basically.

So we tend not to look at open-ended evolutionary dynamics.

I would say that active inference has some leverage to represent that that other framework could not have because of the relationship between constraints as an clinical thing and beliefs as an intentional state.

Yeah.

One thing that makes me think of.

there could be some developments like in message passing algorithms or interoperability of different generative models or category theoretic augmentation of models that are connected to the active inference ontology.

So in that kind of....

We talk about all the time, surprise minimization, things doing as most likely for them to do, paths of least action, Bayesian mechanics.

Then there's a continuity of explanation with the rock just being there and the investor trading with this strategy, an instrumental unifying account that makes relatively minimal claims about the territory that includes the rock and the investing activity.

What did you add in?

What you said, message passing is interesting to me because you have a broad coalition of physicists, mathematicians, scientists, people in academia who basically think there is one language to nature and what we do as scientists is to discover it.

And I don't think that holds water, not only because of an evolution is a thing, which is a sufficient argument actually.

But if you look at message passing, if you look at quantum Bayesianism, if you look at category theory, open formally and conceptually, the possibilities that there are different languages in the world and they may be more or less interoperable and they may be non-reducible to each other.

And then you start to ask questions, I'd say.

I'm not going to go further down the math road.

interested in what determines the fact

that a specific language works in a specific context.

Why is a, I'm not gonna go further down the math road, but you have important compatibility slash inter-operability result between major fields of physics.

I will give a reference by John Baez.

I don't know how you pronounce it.

I never said this out loud, actually.

Baez.

So you have important inter-operability results in fields of physics and formal science.

And we don't have proof, however, that, you know, all possible representation of physics are interoperable.

Operable.

How do you say that? Interoperable.

Interoperable.

Interoperable.

Interoperable.

Interoperable.

Interoperable.

Interoperable.

Okay.

Okay.

So they said it's wrong.

No, it's, it's, it's a great point.

And what's really interesting is like the relationship between Bayes graphs.

First off, Bayes graphs being analytically described, and then computationally implementable is on the order of decades old.

And then as we kind of explored just yesterday in live stream 55.0, the relationship between the Bayes graph and Forney graphs, which enabled message passing to be done procedurally to optimize a Bayes graph, which just brings all of these computational, all these analytical connections, but that wasn't understood until it was.

And that was only slightly over two decades ago.

So then in fact, those kinds of results and developments reflect the, the changing social and scientific niche.

So I don't know, maybe this is a different question, but science as a social activity.

And then I don't know, does that, how does that ground our work or give us more?

What?

So,

about the whole language thing.

I don't know where it's at now, but at least 10 years ago, it seemed that you can have message passing, you can have, well, you can have Bayes and coherence of state of observable that does not rely explicitly on the probabilities, as in a dispersion that is embedded in a specific set space.

You can have Bayes and coherence that pre-exist to the representation in which probability is embedded.

And if there was a way to cross this and message passing, you'd have a pretty straightforward, straightforward road to a straight account of contextuality and social reconstruction.

I say this with scare quotes.

Oh, great.

Great.

No, exciting for quantum context and quantum social and scripts and constraints and norms.

Let's close that question for now, though.

Nice.

Okay.

How can,

how can,

oh, yeah, great question.

Thank you to whoever asked.

And again, video description, you can ask, it'll appear and we'll see if anyone writes more questions during the stream.

How can or does a framework slash theory slash principle of everything articulate with moral, normative, ethical, deontic considerations?

So I think it is addressed in a good amount of detail by an activist, how you relate a theory of biological individuation to a theory of ethics.

I do not have a good grip on this literature.

So you have like, how to say, high level intuition.

I have a negative opinion on that, but this is not like state of the art.

And,

okay, basically, I, I'd,

I'd say this flows well with stoic,

virtue ethics.

Things act according to the nature.

So you might as well work.

So that your nature is good.

And then you have a recursivity in accounting,

what is a good and what is not.

And I do not see a straightforward analytical way to resolve this ambiguity.

I said circularity or being with the both apply.

Yeah.

So I think Evan Thompson has worked in that and may have a much better answer to how we relate theories of being and theories of ethics.

Oh, very loaded, actually.

Yeah.

Yeah.

I guess my sort of short answer would there, there would be for first, just,

we would have to hear a lot more even possible perspectives, but without even hearing them can at least

maybe say that there is an articulation.

And a meter stick as an instrument.

And this kind of goes actually to that question about the base spaces and what is precondition.

So in what ways is the meter stick?

Articulate with downstream ethical frameworks.

What ought to be done?

Or just normative more broadly.

There's downstream decisions that the meter stick is agnostic to.

And then the meter stick is also a social result that is itself downstream of other norms and constraints.

Biotic, abiotic constraints.

Regularities of space.

Time.

I would say there is.

What we can do with a theory of sense making, be it active inference or inactivism is to.

Like just.

Describe.

What we mean by normativity and ethics.

What are the underlying natural processes?

When I say natural, it does not mean by opposition to social.

Those processes are biological and social at the same time.

And that may bring reflexivity or distance

to account of normativity.

That may prove some of them incoherent or ill-advised.

That will not give you a solution to what is ethical

and what is not.

Notably because the most natural way

derive ethics from something like an activism

or something like nativity in France

is that both those frameworks are centered

around biological individuation.

So we may say that biological individuation is good, actually.

That sounds sensible.

But then the question is, biological individuation for whom?

Because you have a right, I guess, or desire will at least to exist.

But so does cancer cells within your body,

who trumps who? I don't know.

Not from a purely scripted account.

There is no way you can straightforwardly,

anonymously derive a prescriptive notion of good
from any physics, any descriptive account of the world.
So you will have, when you make normative claims,
you have goals because you are a living thing.
And because of active inference slash nativism,
we know that cognition is embedded in normativity,
in the autonomy, in the biological activity of the agent.
So the theory that you are framing is in some way motivated,
even if you don't know how or admit that it is.
And then I'd say from there,
it does not quite destroy the project of an objective system of ethics,
but it makes it much harder to defend.
So yeah, I'd say you can derive a lot of negative results
from descriptive theory of, like, I wouldn't say ethics,
because ethics are non-descriptive,
but descriptive theory of how ethics come to be.
I would say that it is hard, if not impossible,
to derive positive results.
If someone tried to do it, I'd like to see what this is.
Wow.
Thank you, Avil.
That last piece was very interesting.
Many results, many...
Many ant colonies with their behavior to model.
Many real situations to understand social constraints and norms in,
instrumentally,
with descriptions that are then in the discussion section of the paper.
And the data are the data, and those are the descriptive aspects.
It's a different kind of data in a slight way.
But there's already enough to describe and interpret
from what would be merely described,
and it is very hard to derive positive results.
First, it's a hard domain to make positive results in.
Not just ones that make people feel good, but by positive, like, positivism.
Like, justified, true assertions about a metaphysical space.
So it's hard to derive positive results about normativity,
and the kind of areas that this person asked.
These are all areas that are...

And it's hard to use a quantitative theory,
which is almost...

It's an abstraction, or it's just one language.

It's hard to use that to make positive results
external to the system.

Harder to make external to the system.

Positive results internal to the system are dime a dozen.

But then to really make that reflexive loop that you described
between the...

What is and what we expect and prefer,
which is actually woven up in an interesting way in ActInf,
rather than two kind of diverged timelines,
like what we, quote, expect to happen,
like a prediction of the likeliest future,
and then a second timeline that's like,
well, what I would really prefer to be doing.

In ActInf, the agent just has one expectation slash preference,
which loads up on the pragmatic value.

But that gives a unified approach
towards one's expectations of how things will be
and what they expect slash prefer them to be.

I would say that at a vibe check level,
ActInf flows well with the idea that empowerment is good, actually.

But make it a formal relation would be quite hard.

You could say that ActInf vibes well with the notion
that everyone knowing their place is good,
which is quite the opposite.

So I don't see an ambiguous relation
between ActInf and any ethical framework I can think of.

I would say, again,
that it flows well with virtue ethics framework
because of the relation between one's action
and one's nature, according to ActInf.
and one's nature, according to ActInf.

Nature being here in a term that is,
like, maybe not the best term,
but, like, I'm literally using a traduction
of a before Christ text.

So, yeah, nature is not that bad.

Just not what we mean

in the context of the 19th, 20th discussions.

Yeah, that's a very complex topic

with definitely a lot of rich work

on, like, our cities and people, nature.

Are humans, animals?

And that sort of

Homo sapiens-type discourse.

And then how does Active Inference frame

the relationship with the niche

in terms of entanglement

or communication

or co-action,

intra-action,

communication

and influence through

blanket?

Where does that go beyond just describing?

It doesn't have to.

I think a sort of base level

in what you referred to here

is, like,

that artifact,

the product of the work,

which is what the work is about,

can be

a jumping-off point

for other directions.

And so the work of Active Inference

Modeling can be

in what it leads to.

It's not always

internal to the model

that the

improved awareness

or better decision

is realized.

I can see one
quite straightforward
and I
don't think it is
entirely broken
way to relate
ethics to Active Inference
is
ethics
like,
not ethics,
normativity
as a social phenomenon.
It is a collective phenomenon.
It's about
social relationships.
And you could
apply
a descriptive
theory
of human behavior
and, like,
human,
social,
aggregated,
collective behavior
to
find out
what traits
in this organization
are prone
to survival
or adaptivity.
And you could say
that those traits
are
good,
actually,

by construction
of what we mean
by ethics.
That could be
a coherent claim.
And then you could
be interested in
what traits
make an organization
durable.
this is the only
logically straightforward
way I see
but that would
tend to
identify as good
things that
I would not see
as good
like
theodolism.
It lived
quite a long time.
Well,
theodolism,
let us say
old school empires
with no claim
to representations
or power
by the,
what do you say,
populace.
And you could
integrate,
you could basically
renormalize
this theory

not around
clock time
but around,
I don't know,
metric of lifetime,
how much
things happen
within the organization.
and then you'd
start having
something that
is less
obviously wrong.
And you could
have
basically
ethics
be
about
optimization
of specific
parameters
that can be
addressed
by a descriptive
framework
but you'd have
to make a lot
of jumps
to get there
and while
I do not
see an obvious
problem to that,
it is very
possible that
we would
not like

the conclusion
of such
optimization
thing
and
again,
if you want
to model
something,
you have to
decide what
are the
parameters
of interest
and you
can be
wrong
about that.
So here
again,
do not have
an ambiguous,
even at a purely
descriptive level,
you do not have
an ambiguous
application
of anything
to anything.
I would say
that more
empowerment
is good
actually
and I would
say this
goes at
many

nesting
scales
of
organization
and I
would say
it has
somehow
something to
do to
the physics
of biological
individuation
but
only in
a vibe
check way.
Nothing
wrong with
that.
Yeah,
especially
with some
of the
recent
work
like the
variational
synthesis
with
Eco,
Evo,
Devo
and the
connections
between
the
statistical

concepts
of
persistence
or
thickness
from
free
energy
principle
related
to
re-measurement
of the
observer
or the
organism
of
himself
that
concept
of
repeated
measurement
being
compatible
in our
work
and in
other
related
works
basically
with
fitness.
If
there's
a height
distribution

of trees
in the
forest
and then
a certain
time
point
later
the
height
distribution
is
the
same
that
was
stabilizing
dynamics.
We could
say
stabilizing
evolution
or
stabilizing
selection
if it
moved
it
moved
and
so
how
do
since
there
is
such
a

coupling
between
things
being
as
they
are
and
basically
what
the
distribution
of what
is
there
is
described
both
this
stoic
way
but
also
this
kind
of
evolutionary
biology
way
so
does
active
inference
offer
any
other
or
complementary

perspective
other than
just
the
kind
of
strengths
and
weaknesses
of
the
of
using
fitness
phenotypic
fitness
or
mimetic
fitness
as
its
own
justification
to
say
that
is
to
say
that
something
is
good
because
it
exists
this
is

not
a
strong
position
to
take
so
this
is
why
you'd
have
quite
high
level
of
difficulty
to
derive
anything
prescriptive
from anything
descriptive
the
function
is
not
the
same
you
can
make
relation
between
the
two
no
ethics

is
purely
blind
about
nature
no
theory
of
nature
is
purely
ethically
unmotivated
but
to
have
formulation
between
the
two
is
probably
not
possible
at least
highly
non-trivial
okay
so to
kind of
close on
this
question
how does
the free
energy
principle
as a

principle
for
everything
and active
inference
as an
implementable
framework
for applying
free energy
principle
systems
how does
that
articulate
with
these
broader
very
deep
and
fundamental
human
social
contexts
short
answer
seems
to
be
it's
a lot
like
other
scientific
ideas
inherits
a lot

of their
challenges
and frictions
and
situations
also
there
appear
to be
some
interesting
promising
routes
or
themes
to weave
in
however
even
with a
very
charitable
or
optimistic
interpretation
there
seem
some
in
principle
challenges
with
these
areas
that
make
certain
kinds

of
claims
perhaps
not
even
within
the
proximate
understanding
we have
of active
inference
but people
should
totally
write
in the
live
chat
or
add
comments
and
stuff
I
mean
it's
just
interesting
question
okay
are
higher
order
beliefs
about
doing
better

than
expected
at
PEM
probably
prediction
error
minimization
integrated
into the
architecture
of the
agent's
internal
model
of
external
dynamics
and
itself
in
those
dynamics
my
thought
here
is
that
we
have
prior
beliefs
to
fulfill
certain
self-evidencing
expectations
e.g.

homeostatic
predictions
but how
do those
interact
with these
higher
order
predictions
which
appear
to my
mind
to deny
or negate
the
fundamental
priors
in so
far as
they
predict
the
agent
to go
over
and above
the
levels
of
prediction
error
minimization
which
otherwise
might be
achieved
by fulfilling

those
fundamental
priors
they gave
their
thought
let's
return
to the
original
question
i do
not know
like i know
that
comparison
to a
baseline
homeostatic
level
of
prediction
error
minimization
is a
quite
important
thing
in
the
computational
psychiatry
was
in
the
work
of
bark

solms
i do
not
know
how
that's
implemented
actually
in
computational
models
so i'd
rather
not
answer
approximately
ben white
should be
more informed
about
these
things
yeah
or maybe
you
well
i mean
the word
that seems
to be
doing
the work
is integrated
so
higher
beliefs
about
doing

better
than
expected
prediction
error
minimization
or we
can just
say
free
energy
minimization
so this
was explored
in the
affective
inference
work
so if
the
free
energy
was
decreasing
as expected
it was
associated
with
neutral
valence
if
things
were
improving
better
than
expected
it was

associated
with
positive
valence
and if
things
were
doing
worse
than
expected
it was
negative
valence
so
because
of the
ability
to make
hierarchical
active
models
and
just
hierarchical
bayesian
models
in general
it is
possible
to talk
about
these
beliefs
about
beliefs
metacognitive
beliefs

so it
is possible
to make
models
that have
expectations
about how
well even
an optimization
process is
going to go
and then
a computational
psychiatrist
or whatever
may associate
that with
the affect
how is that
integrated
into
the agent's
internal
model of
external
dynamics
well
shortest
answer is
to sketch
it out
how one
sees it
and then
it becomes
either
there or
not

but just
in general
if the
internal
model
includes
these
higher order
beliefs about
doing better
than expected
then
that's
the agent's
generative
model
that then
interfaces
or articulates
with external
dynamics
so you
could have
some
environment
that has
an
environmental
regularity
or dynamics
that are
influenceable
or not
and then
you could
deploy two
agents
one of

them
that has
the baseline
generative
model
and the
other one
has this
doing better
than expected
layer
that the
questioner
is asking
about
and then
they can
do
parameter
sweeps
and
maybe
adding
this
higher
order
belief
increases
the
state
space
a lot
and so
it
increases
the
computational
cost

a lot
but some
interesting
pockets
of
behavior
are
found
or
the
behavior
in that
setting
ends up
looking
the exact
same
but in
another
setting
it
would
have
been
different
a lot
of
these
questions
point
towards
very
important
modeling
questions
which can
basically
only be

pointed at
because
so much
is going
to depend
on what
the actual
model
is
okay
um
where can
I find
the
recommended
reading
list
mentioned
in the
lectures
so
where we
have
the
um
slides
from
the
lecture
they're
in
live
streams
just go
to the
big
table
and find

the
slides
if you
find
a
lecture
that
doesn't
have
slides
just
let
us
know
we'll
we'll
add
it
um
where
things
were
just
mentioned
in
the
lectures
that's
a
great
question
and
our
goal
at
the
active
inference

journal
beyond
just
transcribing
and
publishing
and
adding
captions
and
translating
and
everything
of
these
live
streams
we
would
like
to
also
do
more
bibliographic
work
so
that
the
books
that
people
mention
we can
also
have
in that
open

source
versioned
journal
so we
generate the
first version
of the
transcriber
people update
it and prove
it
oh what was
the name of
that author
what book
are they talking
about here
and then we can
actually have a
unified bibliography
section
for the stream
so
that sounds
interesting to
you
and you'd
benefit from
it
then surely
many many others
will too
so
if you're
excited to
do that
kind of
active

inference
journal
work
I highly
encourage
slash
recommend
it
I
about
what else
will you
add
same
I
can have
nothing to
add
yeah
okay
well
we can
look at
some of
the other
questions
that were
added
that others
wrote
people in
live chat
can add
questions
or they
can go
to the
course

website
and
ask a
new
question
do you
want to
look at
one of
the previous
ones
or
do you
have any
other
thoughts
you want
to
bring up
not
really
the
first
ones
were
answered
so I
don't
like
and
they
were
asked
weeks
ago
so there
is a
relatively

low
likelihood
that
going
over
them
again
would
be
very
helpful
I
think
we
can
cut
the
discussion
cool
important
things
that are
not
treated
now
can
be
treated
in
conclusion
anyway
okay
so what
are the
next
steps
then
for

the
course
next
step
is
did
a lot
of
people
register
for
the
presenting
in
the
conclusion
session
I
haven't
gotten
any
specific
responses
or
titles
yet
okay
so we
see
people
do
that
yeah
we
can
write
another
message

including
even
a short
video
message
or
just a
written
one
and then
also
ask
specific
people
also
if
anyone
is
interested
to
present
something
but
they're
not
really
sure
the
format
or
they
want
pre-recorded
we're
just
kind
of
exploring

and
seeing
what
the
people
who
are
following
along
live
want
to
do
and
also
trying
out
things
that
we
think
futures
will
want
to
do
too
so
if
someone
has
an
idea
on
that
we're
definitely
open

to

it

any

last

thoughts

nope

I'm

good

cool

all

right

thank

you

then

see

you

at

all

bye

bye